Sunwoo Hong

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Research Interests

I work on **closing the gap between what people do easily and what models still cannot**: people reason instead of matching patterns, learn from very little, and act in a physical world they understand. So far I have pursued this across **reasoning**, **efficient learning**, and **world models for embodied agents**. My goal is **systems that work outside the lab**: they stay right when a problem is rephrased, a request changes midway, or an object is out of sight, and they get better from their own attempts rather than from more human labels. This will take new algorithms and architectures, and I want to build them.

Keywords: reasoning, world models, embodied AI, RL, diffusion language models, data-efficient learning

Education

Korea Advanced Institute of Science and Technology (KAIST) — Seoul, South Korea
M.S. in Artificial Intelligence, Graduate School of AI — *Feb 2027 (expected)*
Advisor: Prof. Se-Young Yun

Seoul National University (SNU) — Seoul, South Korea
B.S. in Electrical and Computer Engineering — *Feb 2024*

Sejong Academy of Science and Arts — Sejong, South Korea
Inaugural Student Council President; Government-designated academy for gifted students — *Feb 2018*

Publications

∗: equal contribution (random order), †: corresponding author, C: conference, W: workshop, U: under review, P: in preparation, D: domestic

[C2] S. H. Kim∗, **Sunwoo Hong**∗, Y. Choi, C.-H. Chao, S.-Y. Yun†, and R. G. Krishnan†. Pivot-SD: Efficient Self-Distillation for Masked Diffusion Language Models. **Empirical Methods in Natural Language Processing (EMNLP), Main Conference**, 2026. **Oral presentation.** [project]

[C1] S. H. Kim∗, **Sunwoo Hong**∗, H. Jung, Y. Park, and S.-Y. Yun. KLASS: KL-Guided Fast Inference in Masked Diffusion Models. **Neural Information Processing Systems (NeurIPS)**, 2025. **Spotlight (top 3.2% of submissions).** [pdf] [code]

[U1] **Sunwoo Hong**∗, S. H. Kim∗, C.-H. Chao, Y. Choi, S.-Y. Yun†, and R. G. Krishnan†. Logical Equivariance in Diffusion Language Models. Under review at **ICLR 2027**. Earlier version presented at **ICML Workshop** on Foundations of Deep Generative Models, 2026. [pdf]

[P1] **Sunwoo Hong** and S.-Y. Yun†. World Models Need Judgment. In preparation for submission to CVPR 2027.

[P2] **Sunwoo Hong**, A. Pesaranghader†, and S.-Y. Yun†. Neuraxis: From Dual-System to Nervous-System Control in Vision-Language-Action Robots. In preparation for submission to ICML 2027. [draft]

[W1] C. Park, **Sunwoo Hong**, K. Kim, Y. Park, A. W. Olson, and A. Pesaranghader†. Route-Guided CBF-QP Repair for Safe Vision-Language-Action Models. **RSS Workshop** on Trustworthy Embodied Foundation Models, 2026. [pdf] [project] [code]

[D1] **Sunwoo Hong**, Y. Kim, S. Jung, J. Oh, and S.-Y. Yun. Memory Reduction via Ternary Quantization in Reinforcement Learning. Korea Institute of Military Science and Technology (KIMST) Conference, 2025.

Research Experience

Graduate Research Assistant, OSI Lab, KAIST AI — *Feb 2025 – Present*
Advisor: Prof. Se-Young Yun [page]

- Showed world models lack the judgment decisions need: a 2B video world model's futures pick the best push at chance, yet a recorded grasp is judged at AUC 0.85 with 16 labels (0.70 beforehand with 158). Read-to-Verify learns it from practice: 16 labeled episodes nearly match 158, and a vision-language reader needs no labels [P1].
- Built a nervous-system control stack for VLAs, where a planner writes checkable contracts and fast layers check them every action chunk. Proved that standard data lets scene and posture override the instruction [P2].
- Found that the KL divergence between consecutive denoising steps tells a masked diffusion model which tokens are already settled. Unmasking those in parallel, with no training, gives up to 2.78× faster generation while improving on greedy decoding [C1].

Visiting Researcher, University of Toronto & Vector Institute *Jan 2026 – Jul 2026*
Advisor: Prof. Rahul G. Krishnan [page]

- Showed that diffusion language models solve about 99% of Sudoku states in at least one of eight rotated or reflected views but only 13–24% in all eight, and proved that hard-wiring this symmetry into attention leaves it almost no capacity, so consistency must be learned [U1].
- Found that a few early commitments decide most of a diffusion language model's output. Training on those alone (10 tokens per trajectory, under 4%) beats full-sequence fine-tuning and RL runs with ten times the data, at one-sixth to one-eighth of the wall-clock time [C2].

Research Collaborator, LG Electronics Toronto AI Lab *Jan 2026 – Jul 2026*
Mentor: Dr. Ali Pesaranghader [page]

- Safety filters keep a robot policy from colliding but push it off its task path. Co-developed a module that routes it around the obstacle and back, improving collision avoidance and task success over AEGIS on SafeLIBERO [W1].

Graduate Research Assistant, MLML Lab, KAIST School of Computing *Feb 2024 – Jan 2025*
Advisor: Prof. Sungjin Ahn [page]

- Designed a compact 1D tokenizer that compresses video into a short sequence of global tokens instead of a spatial grid, to model scene-level dynamics in video world models.
- Studied why block-slot object-centric models fail to separate factors such as shape and color, and built out-of-distribution tests that measure compositional generalization.

Undergraduate Researcher, AISys Lab, SNU *Jan 2023 – May 2023*
Advisor: Prof. Jinho Lee (B.S. thesis) [page]

- Built pipeline-parallel training schedules for knowledge distillation that remove idle time between teacher and student passes: up to 1.45× faster training or 1.55× better memory efficiency than a GPipe baseline.

Research Intern, Human-Centered Computer Systems Lab, SNU *Jun 2022 – Sep 2022*
Advisor: Prof. Youngki Lee [page]

- Built a server-driven adaptive video streaming pipeline for deep learning inference, reducing bandwidth by up to 70% while maintaining detection accuracy.

Undergraduate Research Program (URP), SNU *May 2021 – Dec 2021*
Advisor: Prof. Kyomin Jung [page]

- Developed a GAN-based model that transfers a sign-language interpreter's motion, extracted as skeletons, onto the speaker in a video, so that the speaker appears to sign.

Awards and Honors

- Oral Presentation (2.9% of submissions), EMNLP 2026 *Oct 2026*
- Excellence Selection, GARL 2026 Challenge (National AI Research Lab, Korea) [U1] *Jul 2026*
- Spotlight (top 3.2% of submissions), NeurIPS 2025 *Dec 2025*
- Full Scholarship, Choi Hyuk-Young Scholarship Foundation *Mar 2019 – Present*
- Minister of Education Award (1st place), national STEAM R&E research competition *Nov 2015*

Teaching and Academic Service

- Teaching Assistant, Geometric Deep Learning (AI810), KAIST *Spring 2025*
- Teaching Assistant, Reinforcement Learning (CS672), KAIST *Fall 2024*
- Teaching Assistant, Freshman Seminar: Writing (E51.101), Seoul National University *Fall 2021*
- Reviewer, ICLR 2027 and ACL Rolling Review (August 2026 cycle) *2026*

Military Service

- Republic of Korea Air Force, Sergeant (honorable discharge); information systems specialist *Aug 2019 – May 2021*